ME EN 7210 Optimal Controls **Spring 2026**
Assignment #5 **Due Tuesday, Mar. 3 at the beginning of class**

Problem 1 (Exact system inversion problem)

(a) Consider the following piezo-base positioning system:

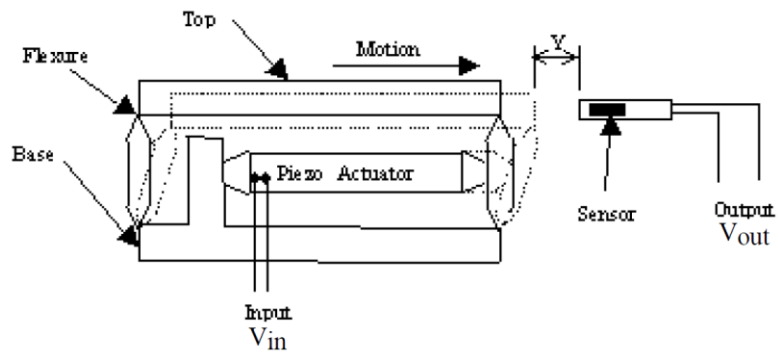


Figure 1: Schematic of piezo-stage

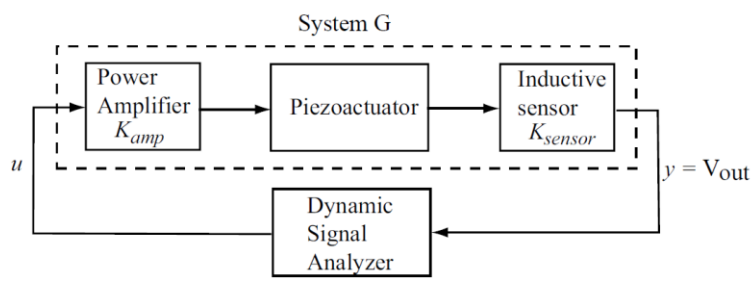


Figure 2: Modeling the vibrational dynamics using a dynamic signal analyzer

The piezo-stage is a positioning system that can be used in applications that require small (tens of micrometers) displacements. The piezo-stage consists of a piezo-actuator, flexural mechanism and a sensor as shown in Figure 1. An input voltage is applied to a piezo-actuator, which expands and pushes against the flexural mechanism that amplifies the piezo-movement. The displacement of the piezo-stage is measured using and inductive sensor. The inductive sensor has a gain of 0.2 volts/µm and measures a output voltage (VOUT). Figure 2 shows the actual system itself connected to an instrument called a Dynamic Signal Analyzer to measure the system's frequency response (Bode plot), which is then used to determine a transfer function. Afterwards, the transfer function is converted to a state-space model (in control canonical form) given by:

$$\dot{x} = \begin{bmatrix} -1.169 & -50.3 & -45.94 & -685.3 & -391.7 & -1952 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 11.88 & 4.977 & 539.6 & 129.9 & 5625 \end{bmatrix} x$$

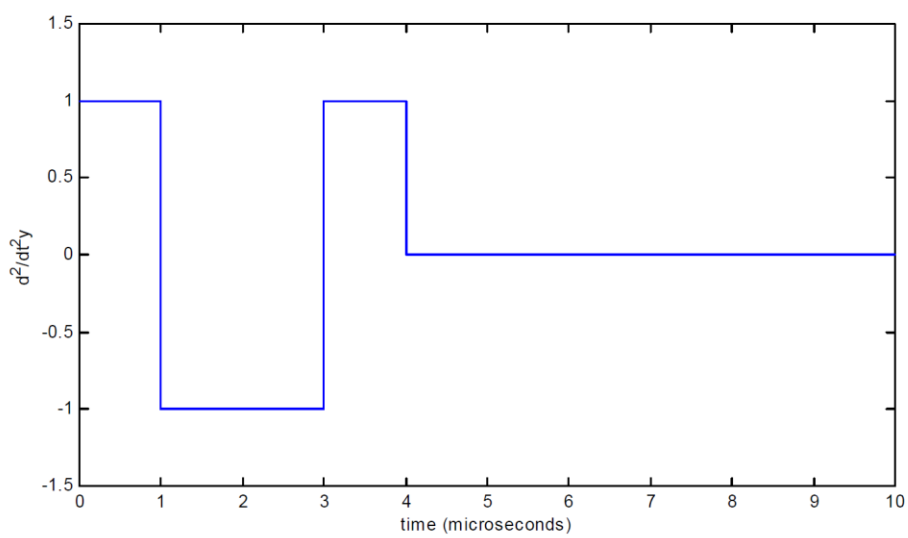
$$\dot{x} = Ax + Bu$$

$$y = Cx$$

Determine the relative degree of the system. Note that relative degree means the difference in the order between the numerator and denominator polynomial of the system transfer function.

(b) Determine the state-space model of the inverse system given some desired output trajectory $y_d(t)$ that you want to track using the system inversion technique. What are the eigenvalues of the inverse system and how do they compare to the eigenvalues of the original system? You may want to compare the poles of the original system transfer function to the inverse system transfer function to gain some additional insight.

(c) Simulate the response of the inverse system model using MATLAB (lsim command) to find the inverse feedforward input. The desired acceleration profile is shown below.



In your simulation, show plots of the $y_d(t)$ and its first time derivative, and also show the response of the inverse system, that is the behavior of the inverse states (reference states per the notes in class).

(d) From the results above, determine the feedforward input and show plots of the input vs. time. Apply this feedforward input to the system model given above to verify that output

tracking is achieved (use lsim command again) --- do not use state feedback! Show plots of the desired output $y_d(t)$ compared to the output of the system when you applied the inverse input you determined. Does the system track the desired output using the inverse feedforward input?

Problem 2 (Optimal system inversion proof)

In class, Dr. Leang lectured on the frequency-based optimal inversion process and gave the essential equations for the optimal input and optimal output trajectories that minimize the cost function:

$$J(U) = \int_{-\infty}^{\infty} \left[U^*(j\omega)R(j\omega)U(j\omega) + e^*(j\omega)Q(j\omega)e(j\omega)\right] d\omega,$$

where `*' denotes the complex conjugate transpose, $U(j\omega)$ is the feedforward input to the system, $e(j\omega) = Y(j\omega) - Yd(j\omega)$ is the tracking error, $Y(j\omega)$ is the output, $Yd(j\omega)$ is the desired output, and $R(j\omega)$ and $Q(j\omega)$ are non-negative, frequency-dependent real-value weights on the input energy and the tracking error, respectively.

Show that for an open-loop system with model $G(j\omega)$ the optimal input and output are given by:

$$U_{opt}(j\omega) = \left[\frac{G^*(j\omega)Q(j\omega)}{R(j\omega) + G^*(j\omega)Q(j\omega)G(j\omega)}\right] Y_d(j\omega).$$

$$\begin{aligned} Y_{opt}(j\omega) &= G(j\omega)u_{opt}(j\omega), \\ &= G(j\omega)\left[\frac{G^*(j\omega)Q(j\omega)}{R(j\omega) + G^*(j\omega)Q(j\omega)G(j\omega)}\right] Y_d(j\omega), \\ &\triangleq G_f(j\omega)Y_d(j\omega). \end{aligned}$$

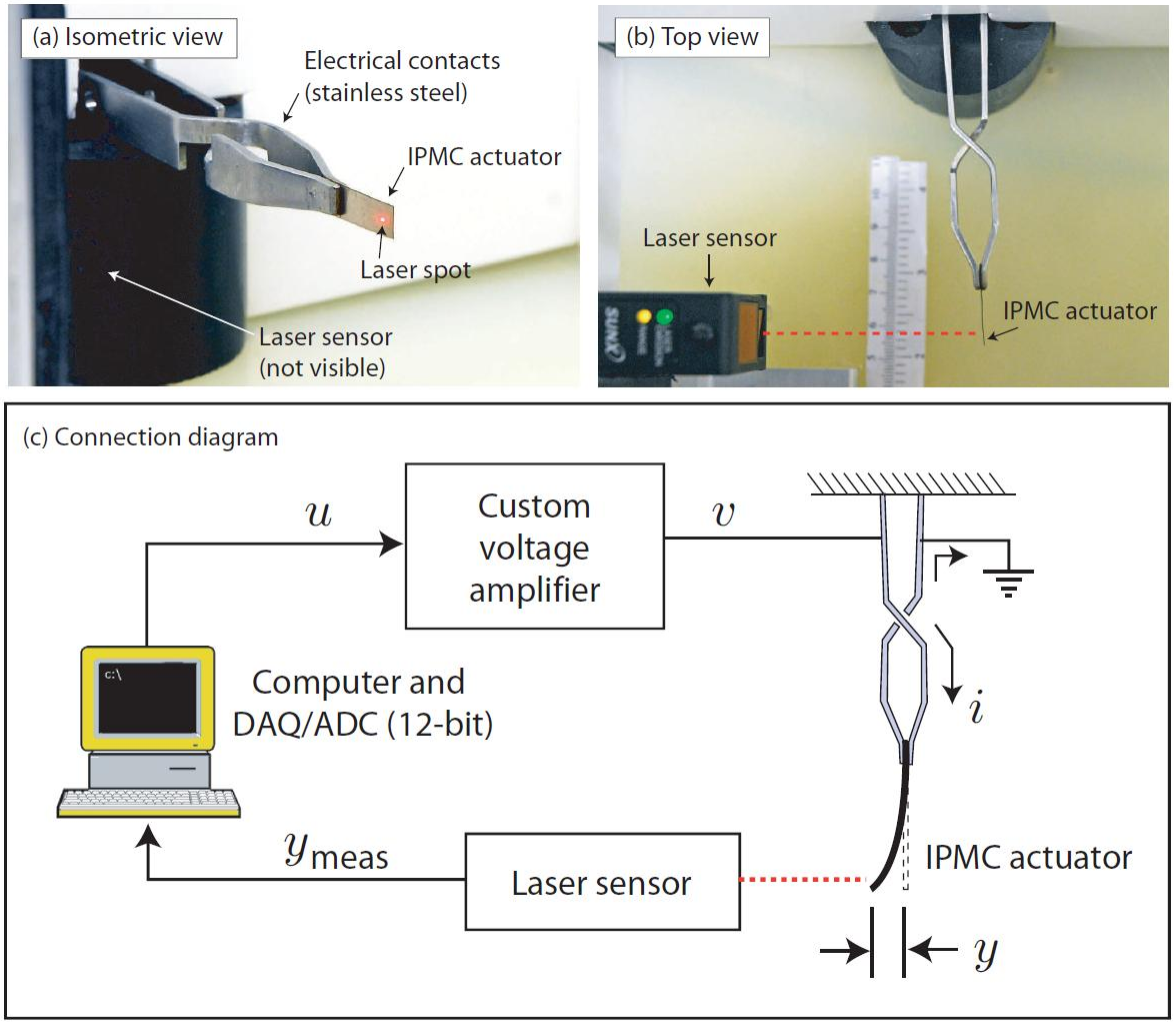
Hint: compute and solve the following:

$$\frac{dJ(U)}{dU} = 0,$$
$$\frac{d^2J(U)}{dU^2} > 0.$$

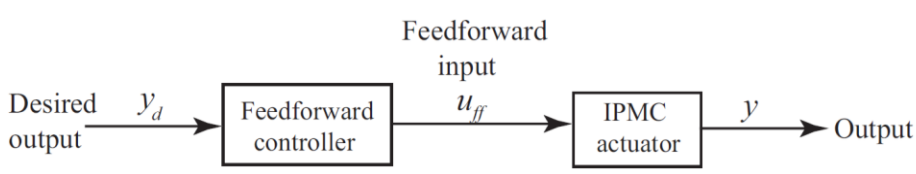
Problem 3 (Optimal system inversion implementation on IPMC actuator system)

In this problem, you will implement the frequency-based optimal inversion approach to feedforward control an example smart-material actuator, specifically an ionic-polymer metal

composite (IPMC) actuator. An IPMC actuator is a type of smart material where an input voltage causes it to displace, creating motion:



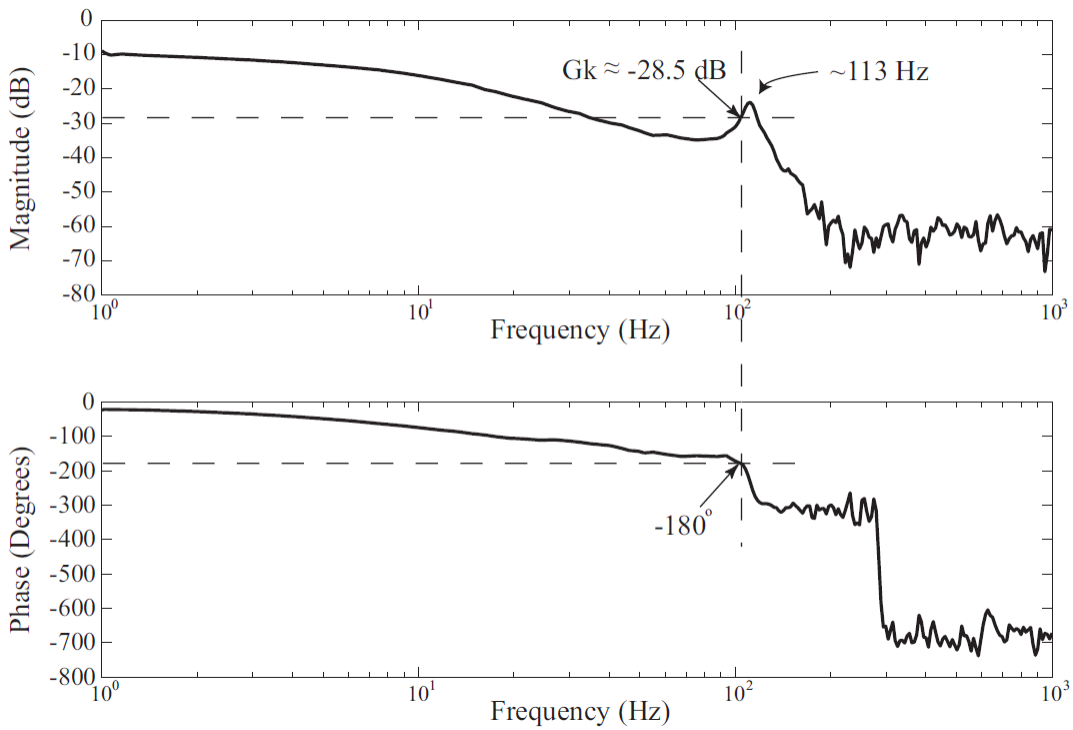
The control diagram is shown below where your job is to final the optimal feedforward input u_{opt}, which is essentially a modified version of the exact feedforward input u_{ff}.



The equations you'll implement is given above in Problem 2 and the optimal input and output will minimize the cost function:

$$J(U) = \int_{-\infty}^{\infty} \left[U^*(j\omega)R(j\omega)U(j\omega) + e^*(j\omega)Q(j\omega)e(j\omega)\right] d\omega,$$

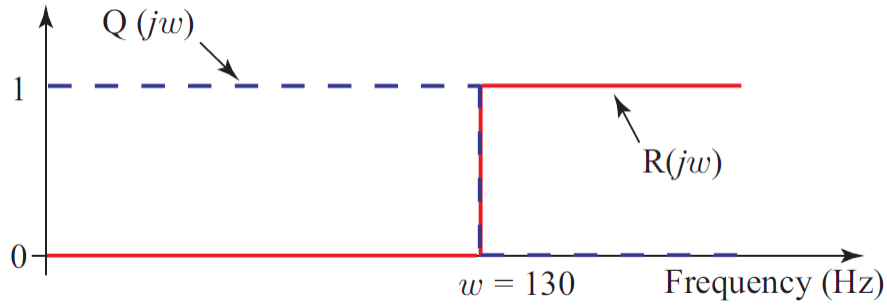
The model of the IPMC actuator was obtained experimentally using a dynamic signal analyzer that measures the system's Bode plot (frequency response):



The frequency response plots for $G(j\omega)$ of the IPMC actuator is shown above, where the Gk is the gain margin.

For the optimal inversion process, you will use $G(j\omega)$ above, along with using a triangular trajectory with different frequencies: 1Hz, 5Hz, 10Hz, and 18Hz. You will determine optimal inputs and outputs for these different trajectories and compare the results.

Pick the weights as follows: $Q = 1$ and $R = 0$ for $\omega \in [0; 130]$ Hz and $Q = 0$ and $R = 1$ for $\omega > 130$ Hz as shown below. The weights were chosen such that the computed feedforward input magnitude was less than 5 V.

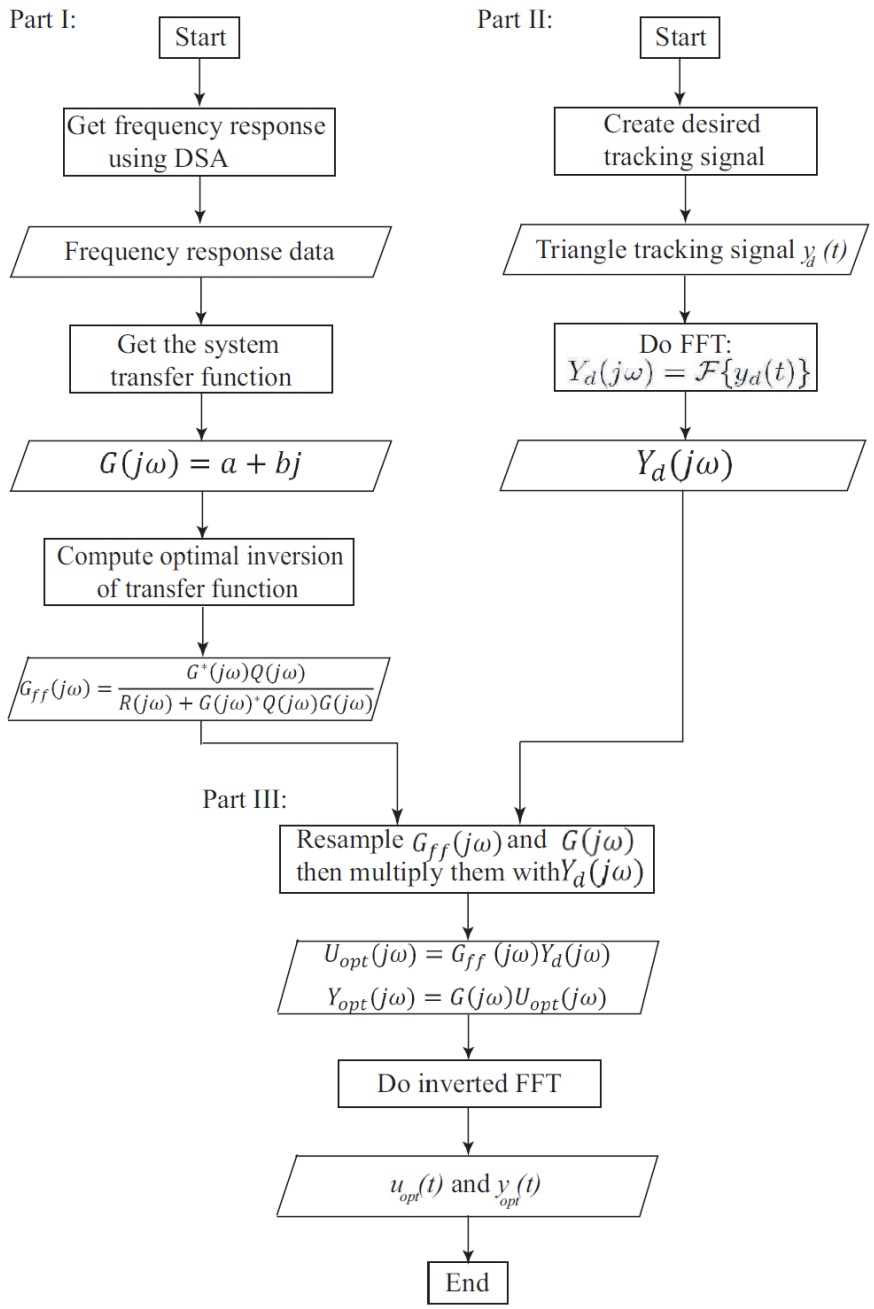


Instructions: Download and extract the Matlab M-file with supporting data files, filename "OptimalInvStart.zip". The extracted files contains the following files:

- Main.m – main m-file that loads the desired trajectory, which is a triangle signal of 1 Hz. Other frequencies are included, such as "yd05.in" for 5Hz, "yd10.in" for 10 Hz, "yd18.in" for 18 Hz. The main.m file shows an example for 1Hz, so you will change the load command to load trajectories for other frequencies (see details below).
- The files "cm.txt" and 'cp.txt" are the magnitude and phase data for the system model, and "cm.x: is the file for the frequency range. The main.m file when you run it will generate the plots for the desired trajectory and the frequency-response model G(jω). You will use these plots in creating the optimal inversion input and output.
- The process to implement the frequency-based optimal inversion process is outlined below, so follow the steps. In Part I, the steps "Get frequency response data using DSA", "Frequency response data", "Get the system transfer function", and "G(jω) = a+bj" is already done in the main.m file. Also in Part II, "Create desired tracking signal" and "Triangle tracking signal $y_d(t)$" is also already done in the main.m file, so please look it over so you understand what it's doing. Your job is to perform the rest of the steps in Part I, II and III. Follow the flow diagram as you will need to implement the FFT on the desired triangular trajectory $y_d(t)$ to get $Y_d(j\omega)$, then compute $U_{opt}(j\omega)$ and $Y_{opt}(j\omega)$, followed by an inverse FFT to find $u_{opt}(t)$ and $y_{opt}(t)$.

Show:

- Plots of the optimal input and output for the 1 Hz triangular signal. Because the trajectory has many cycles, only plot 2-3 cycles when the behavior reaches steady state. Compare the output with the actual desired trajectory, so plot $y_d(t)$ and $y_{opt}(t)$ together, as well as the optimal input $u_{opt}(t)$. Note that you need to compute $u_{opt}(t)$ and apply it to your system model to verify that it's working correctly.
- Do the same for 5 Hz, 10 Hz, and 18 Hz. Compare the results and describe what is happening as the frequency increases. Relate back to the choice of the Q and R weightings as a function of frequency.
- Submit your Matlab code.



The process for the feedforward input computation. Part I was used to treat the model $G(j\omega)$ of the IPMC actuator system and to find the optimal inversion from the system model $Gff(j\omega)$. This part was based on the $Q(j\omega)$ and $R(j\omega)$ weights in frequency domain. Part II was used to treat the given desired tracking trajectory $yd(t)$, *i.e.*, to convert to the frequency domain $Yd(j\omega)$. Finally, in Part III the frequency response function $G(j\omega)$ and $Gff(j\omega)$ from Part I along with the $Yd(j\omega)$ from Part II were used to find the time domain signal $u_{opt}(t)$ and $y_{opt}(t)$.